Interactive lung tumor segmentation using denoising diffusion models

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ABSTRACT

Denoising diffusion models (DDMs) have demonstrated strong generative capabilities for realistic image generation, and their ability to encode uncertainty has motivated applications for medical imaging such as tumor segmentation. Training and inference of denoising diffusion probabilistic models (DDPMs) on whole 3D medical scans is memory-intensive and computationally slow. Prior studies have addressed this challenge using accelerated sampling strategies such as denoising diffusion implicit models (DDIM) and latent space compression approaches such as latent diffusion models, demonstrating improvements in computational efficiency. However, these methods still exhibit high inference times for full 3D volumes, posing a significant challenge for time-sensitive clinical workflows where rapid turnaround is essential. We have developed a novel human-assisted DDM approach (HA-DDM) to address this problem that integrates human guidance for interactive and fast medical image segmentation. Our method takes a user-defined point approximately at the center of the structure to be segmented as an input prompt and crops the image scan to a 3D bounding box region of interest (ROI) of a pre-determined size around the structure, centered at the point. This is fed to the DDPM and DDIM to obtain the output 3D segmentation, which is pre-trained using bounding box ROIs of similar size around the tumor extracted from the training data. We sample multiple segmentations using the DDPM and DDIM to generate spatial uncertainty maps. These segmentations are then ensembled to create the results which are resampled into the full 3D scan space. Our approach improves memory and computational efficiency of DDMs by processing smaller 3D bounding boxes instead of full 3D scans. In this work, we demonstrated our approach for lung tumor segmentation on 2D slices and 3D ROI bounding boxes. We validated our method on held-out test sets, achieving median Dice scores of 0.73 and 0.92 on the Medical Decathlon and institutional 2D datasets respectively using DDPM. For 3D volumes, we achieved Dice scores of 0.77 using DDIM on the institutional dataset with just 20 denoising steps instead of the standard 1000 in DDPM, significantly reducing computational time and resources without sacrificing segmentation quality.

Keywords: Denoising Diffusion Models, Tumor Segmentation, Denoising Diffusion Probabilistic Models (DDPM), Denoising Diffusion Implicit Models (DDIM), Human-Assistance.

1. INTRODUCTION

3D tumor segmentation is a central problem in medical imaging, especially because lesion boundaries can be ambiguous, vary substantially across patients, and are subject to radiologist interpretation. Manual segmentation of all tumors in each scan is labor intensive and difficult to scale across large patient cohorts. Models such as nnU-Net[1] and transformer-based architectures[2] have shown strong baseline performance for automated tumor segmentation. However, they still operate as deterministic predictors, producing a single output segmentation mask without accounting for predictive uncertainty or spatial variability across the mask. In clinical practice, both model confidence and inter reader variability can be captured using segmentation uncertainty.[3] Quantitative assessment of uncertainty of segmentation can be useful (a) to provide confidence measure regarding the segmentation, (b) identify less trustworthy regions within the segmentation, (c) improve trustworthiness for clinical use including treatment response assessment,[4] prediction of treatment planning and response,[5] and patient prognosis.

Recent image generative models, such as denoising diffusion models,[6,7] have shown great promise in addressing this problem. These models have two phases: 1) a forward process, in which Gaussian noise is added gradually to an image across successive time steps until it becomes pure noise; and 2) a reverse process, in which the neural network learns to iteratively remove the noise. During inference, generation begins from a randomly sampled Gaussian noise tensor, and the learned reverse process is applied step by step to produce a coherent image. Unlike deterministic neural networks, diffusion models are inherently stochastic samplers, which can be used to quantify uncertainty and generate multiple

plausible segmentation hypotheses. Prior works, such as diffusion models for implicit image segmentation ensembles,[8] have demonstrated that sampling-based ensembles can approximate inter-observer variability and provide more comprehensive uncertainty estimates than a single deterministic mask.

However, the direct application of diffusion models for 3D tumor segmentation can be computationally challenging. DDIM[7]/DDPM[6] can have high computational requirements and are extremely slow, making them difficult to deploy in real-time clinical workflows for high-resolution imaging. Patch-based approaches, such as PatchDDM,[9] reduce memory demands by operating on local 3D patches, enabling diffusion models to be applied to large medical volumes without sacrificing spatial detail. Despite these advances, developing computationally efficient diffusion-based methods for 3D tumor segmentation tasks while providing meaningful uncertainty estimates remains an active area of research.

We propose HA-DDM (Human-Assisted Denoising Diffusion Model), a method that builds on DDPM and DDIM to enable computationally efficient 3D tumor segmentation. By combining ensemble sampling with a human-assisted region of interest (ROI), HA-DDM generates multiple segmentation masks from different noise initializations, producing an uncertainty map that supports clinical interpretation and adoption.

2. METHODS

2.1 Dataset

We evaluate our approach on both 2D and 3D data sets from two sources: (1) Medical Segmentation Decathlon (MedD),[10] a widely used medical image segmentation dataset. We used the publicly available 63 thoracic CT scans with expert annotated segmentation masks for our experiments. (2) Institutional dataset (Inst.): data set from our institution consisting of 121 cases with lung CT scans and tumor segmentations. The median scanning resolution was 0.69 mm along x and y and slice thickness of 1.25 mm. The ground truth tumor segmentations were performed by a clinical fellow and a board-certified radiologist. The 2D dataset was created from the 3D data set by taking the middle slice of the tumor from the 3D volume.

All CT volumes were resampled to a uniform isotropic voxel spacing of $1 \times 1 \times 1$ mm^3 using nnU-Net's[1] internal resampling pipeline to ensure consistent spatial resolution across datasets. CT image intensities were preprocessed using the DICOM-standard Value of Interest Look-Up Table (VOI LUT) to map raw Hounsfield units to a clinically relevant display range.[11]

2.2 Denoising Diffusion Models (DDM)

We employ denoising diffusion models, including denoising diffusion probabilistic models (DDPM) and denoising diffusion implicit models (DDIM), for 2D and 3D tumor segmentation with uncertainty.

Denoising Diffusion Probabilistic Models (DDPMs)[6] are a class of generative models that learn to produce data by reversing a gradual noising process. During training, Gaussian noise is incrementally added to an input image over a fixed number of time steps until it is fully corrupted into pure noise. The model then learns to reverse this process by estimating and removing the noise at each step, ultimately recovering the original image. For segmentation tasks, the model is conditioned on the input image and iteratively denoises a noise-corrupted mask to produce a final segmentation. While DDPMs have demonstrated strong performance across various image generation and segmentation tasks, the sequential reverse diffusion process results in high computational cost during inference.

Denoising Diffusion Implicit Models (DDIMs)[7] were introduced as a more efficient alternative to DDPMs by reformulating the reverse diffusion process as a non-Markovian, deterministic sampling procedure. Unlike DDPMs, which require traversing the full diffusion chain step by step, DDIMs allow for a reduced subset of time steps during inference while leveraging a model trained on the full diffusion schedule. This enables significantly faster sampling with minimal loss in segmentation quality. In the context of segmentation, DDIMs can generate predictions in far fewer steps, making them more practical for clinical deployment where inference speed is a critical constraint.

2.3 Human Assisted DDM (HA-DDM)

The goal of our work is to adapt DDMs for 3D tumor segmentation tasks that overcomes the accuracy and computation efficiency challenges of DDPMs and DDIMs. We introduce the human-assisted DDM (HA-DDM), an interactive method for generating accurate and real-time 3D segmentations. The key idea of HA-DDM is to generate a tumor segmentation using a single user click representing the approximate center of the ROI, i.e., the tumor in our case that

needs to be segmented. Using the clicked point, 3D bounding boxes of a pre-determined size much smaller than the full 3D scan, are extracted around the tumor to focus on the ROI and fed as input to the DDM. This enables generation of high-quality tumor segmentation masks while also significantly reducing inference time and memory overhead. The user click represents a lightweight interaction that can be naturally integrated into clinical workflows. Unlike conventional deterministic segmentation models, the proposed method generates multiple plausible segmentation hypotheses to estimate uncertainty, thereby improving clinical confidence in model predictions.

2.4 Model Training and Evaluation

The extracted 3D ROI bounding boxes around the clicked point were of a fixed size of $50 \times 50 \times 50$ mm^3, which we found to fully encompass all tumors in our dataset with sufficient background context to aid segmentation. Importantly, some extracted bounding boxes contained multiple tumors, and the model was explicitly trained to handle such cases by combining their ground truth segmentations. During training, click locations were simulated using ground-truth tumor centroids and ROI bounding boxes extracted. During inference, the user clicks generated ROI bounding boxes were passed through the trained DDM models to produce the final segmentation output.

The 3 datasets were split into train, validation and held-out test sets in the following proportion for all experiments: 1) (1) MedD: Train/Validation/Test: 73%/11%/16%; (2) Inst. data: Train/Validation/Test/: 57%/10%/33%); and (3) combined MedD and Inst. data: Train/Validation/Test: 75%/10%/15%. We followed the training procedure in Wolleb et al.[8] for all DDPM training and Bieder et al.[9] for all DDIM training. To generate ensembled results, predictions from multiple runs with different noise initializations were aggregated to produce the final segmentation mask.

2D Models

For 2D images, we evaluated DDPM on all three data sets. The models were trained using the DDPM framework with 1000 diffusion time steps. The final segmentation mask was obtained by aggregating predictions from 5 ensemble runs. To assess cross-domain generalization, we performed cross-dataset inference: the Medical Decathlon-trained model was tested on institutional data, and vice versa. These experiments evaluate generalization across different acquisition protocols without fine-tuning to specific data set. For comparison, deterministic nnU-Net models were trained using default configuration on both MedD and Inst. data sets.

3D Models

The 1000 time step diffusion process of DDPM poses significant computational challenges for both training and inference on high-resolution 3D volumes. We adopted DDIMs, which enable deterministic sampling with significantly fewer steps while preserving image generation quality, making it practical for volumetric medical image segmentation. All 3D experiments were performed on the institutional dataset. We investigated the tradeoff between segmentation quality and computational cost by varying: (1) number of DDIM denoising steps (20, 50, 100, 1000), and (2) ensemble size for the final segmentation output.

Performance Evaluation

We evaluated segmentation performance using the Dice similarity coefficient (DSC),[12] which measures the spatial overlap between the predicted and ground truth segmentation masks. Ensembling of model predictions was performed by averaging the raw model outputs across different noise initialization during inference prior to thresholding. The resulting mean probability map was then binarized at a threshold of 0.5 to produce the final segmentation for DSC computation. For variance maps, the per-voxel variance across all ensembles is computed and reported directly, without any thresholding. Finally, we evaluated inference time for both the 2D and 3D models, with all reported values corresponding to single predictions.

3. RESULTS

2D Results

Figure 1 shows Dice score boxplots across all 2D experiments. DDPM models trained on in-domain data achieved strong segmentation performance across all datasets. Mean and median Dice scores were 0.75 and 0.74 for the Medical Decathlon dataset, 0.87 and 0.90 for the institutional dataset, and 0.84 and 0.88 for the combined dataset. The institutional dataset exhibited superior performance, likely due to more consistent imaging protocols and higher data quality.

Cross-domain generalization revealed substantial performance degradation. The Medical Decathlon trained model tested on Institutional data achieved a mean Dice of 0.49 (median: 0.56), while the institutional data trained model tested on Medical Decathlon data achieved a mean Dice of 0.40 (median: 0.41). These results indicate limited cross-domain generalization, highlighting the key problem of data distribution shift across institutions, scanners, and population.

nnU-Net established competitive baselines with mean and median Dice scores of 0.78 and 0.86 on Medical Decathlon data and 0.93 and 0.94 on the institutional data. Importantly, while nnU-Net outperformed DDPM on both the Medical Decathlon and institutional datasets, DDPM achieved competitive segmentation performance while additionally providing prediction uncertainty estimates unavailable in deterministic models.

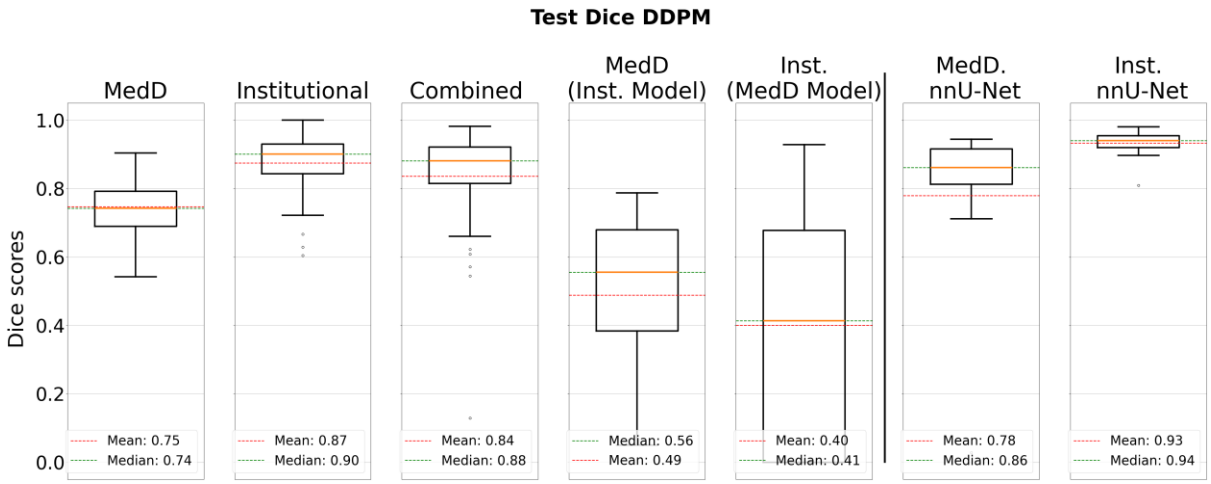


Figure 1: Dice score boxplots comparing within-dataset and cross-dataset generalization performance of DDPM across medical decathlon (MedD), institutional, and combined data sets and baseline nnU-Net.

3D Results:

For 3D data, we focused on two key aspects of DDIM performance. First, we assessed tumor segmentation accuracy. Second, we investigated the tradeoff between computational efficiency and segmentation quality by analyzing output dice scores as a function of DDIM sampling steps and output ensemble size. We identified the minimum number of sampling steps required to achieve comparable performance as the full 1000 denoising steps.

Our HA-DDM approach achieved a mean dice score of 0.77 (median: 0.81) on the held-out test set using DDIM for inference, while nnU-Net demonstrated better performance with a mean dice of 0.84 (median: 0.88), as shown in Figure 2.

DDIM based sampling provides the flexibility to optimize the number of denoising steps, in turn, reducing the inference time without sacrificing segmentation accuracy. We evaluated this trade-off by computing dice scores across a range of sampling steps and ensemble sizes, as shown in Figure 3. This experiment had two objectives: first, to determine minimum number of steps required to maintain segmentation quality; and second, to assess whether ensemble size influences the final segmentation performance. As seen in the heatmap, our analysis indicates that 20 denoising steps with an ensemble size of 3 achieves the same performance as the complete 1000 denoising steps with much lower computational cost.

We also examined the computational time gains achieved through reduced denoising steps and going from full scan to ROI using our HA-DDM approach for segmentation. Figure 4 shows the inference time boxplots across DDIM sampling steps. Inference time scaled approximately linearly with the number of DDIM sampling steps, with mean inference times of 0.90s, 2.24s, 4.48s, and 44.73s for 20, 50, 100, and 1000 steps respectively. DDIM with 20 steps achieved a 49.7× speedup compared to 1000-step sampling, demonstrating substantial computational savings without sacrificing segmentation quality.

Figure 5 demonstrates the computational benefit of our HA-DDM approach over full size image processing using DDMs for both 2D and 3D images. For 2D evaluation, DDIM with 1000 denoising steps required a mean inference time of 38.58s (median: 37.78s) for a 50×50 ROI using HA-DDM, compared to 71.33s (median: 66.44s) for a full 256×256

slice, representing an approximately 1.85-fold reduction in inference time. For 3D evaluation, DDIM with 20 denoising steps required a mean (median) inference time of 0.90 (0.90)s for the HA-DDM ROI of 50×50×50 and 5.28 (5.28)s for the full scan of size 128×128×128, representing an approximately 5.9-fold reduction in inference time. Notably, despite the higher computational complexity of 3D segmentation, DDIM with 20 denoising steps on 3D volumes was 42.9× faster than DDPM with 1000 steps on 2D slices at comparable resolution.

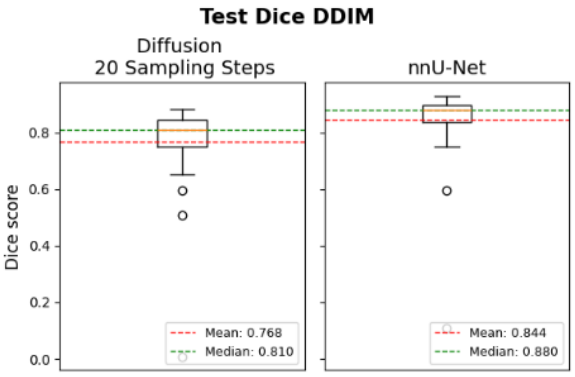


Figure 2: DDIM vs nnU-Net for 3D tumor segmentation using HA-DDM approach.

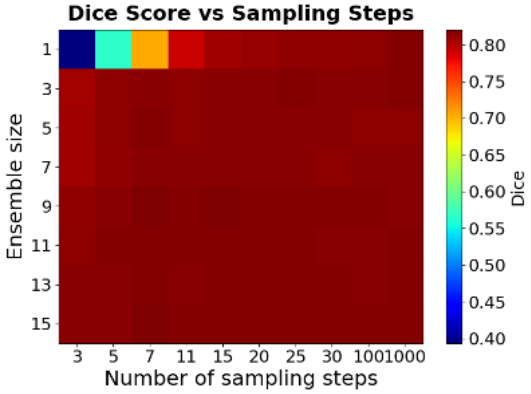


Figure 3: Heatmap showing the trade-off between ensemble size and DDIM sampling steps and their effect on segmentation performance.

4. DISCUSSION

This study presents HA-DDM, a human-assisted denoising diffusion model methodology for lung tumor segmentation in CT scans that provides uncertainty quantification while mitigating the computational limitations of standard DDM-based approaches. By incorporating a single user-clicked point as input to define a region of interest (ROI), the method crops the full 3D scan to a fixed 50×50×50 mm^3 bounding box around the tumor, significantly reducing memory usage and inference time compared to full-volume processing. We explored both DDPM and DDIM formulations, achieving median Dice scores of 0.73 and 0.92 on Medical Decathlon and institutional 2D datasets, and 0.77 on institutional 3D dataset.

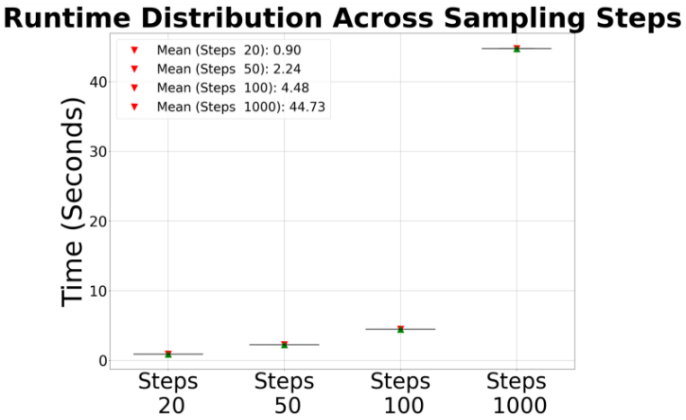


Figure 4: HA-DDM inference times for different number of DDIM sampling steps.

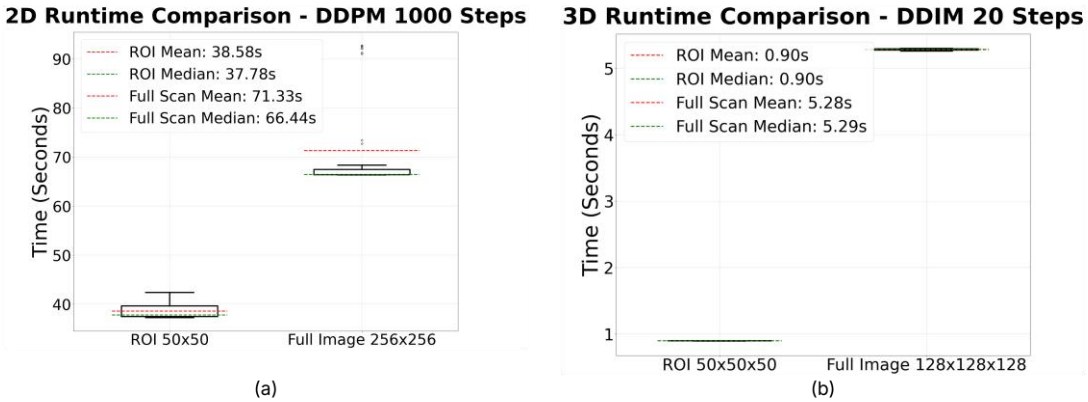


Figure 5: HA-DDM ROI vs full scan inference times for (a) 2D using DDPM and (b) 3D using DDIM.

Computational Efficiency: Sampling Steps and ROI-Based Inference

A key contribution of this work is demonstrating that diffusion-based segmentation can be made computationally practical through a combination of human-assisted ROI extraction and reduced DDIM sampling steps. Our results demonstrate that restricting training and inference to a smaller-sized ROI centered on a user-provided click significantly reduces memory usage and inference time compared to full-volume processing. Furthermore, segmentation performance plateaus at a small number of DDIM denoising steps, with as few as 20 steps achieving Dice scores comparable to those of the full 1000-step process. For 3D segmentation specifically, this translates to a 42.9× speedup, enabling clinically practical inference times without sacrificing accuracy. Together, these strategies make denoising diffusion models feasible for real-time or interactive clinical workflows.

Ensembling and Uncertainty Quantification

Although Dice scores plateaued beyond a small ensemble size, ensembling remains essential for generating voxel-wise mean and variance maps that provide spatially localized uncertainty estimates, as shown in Figure 6. These uncertainty maps offer clinically relevant information by highlighting ambiguous tumor boundaries and regions of disagreement across samples, enhancing interpretability, reliability, and clinical trust of segmentation output. Although nnU-Net continues to outperform diffusion models, deterministic models lack the ability to provide output uncertainty maps enabled by DDM-based approaches.

In conclusion, HA-DDM establishes a computationally efficient, uncertainty-aware framework for 3D lung tumor segmentation in CT scans, bridging the gap between the generative power of diffusion models and the practical demands of clinical deployment.

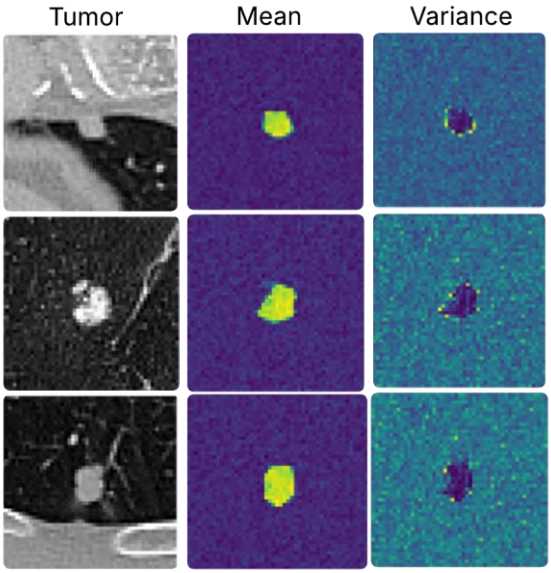


Figure 6: Ensemble mean and variance maps for tumor segmentation using our HA-DDM approach.

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